

Do you provide sourcing or R&D support for startups and academic institutions?

Yes. Several startups have designed their own boards based on the THEJAS 32, using our reference designs as a starting point. They can add, remove, or modify features, including the form factor, according to their ideas. Typically, they create the schematic and share it with us for review. We provide guidance through online meetings, advice on necessary changes, and support them throughout board design and testing.

For IP development, we provide VEGA-based FPGA designs to industry and academia, enabling them to implement and test them. We also host hackathons and challenges where users can integrate their own IP into VEGA-based designs. Over the past two years, we have collaborated with the annual Design & Verification Conference (DVCon) to support its design contest using VEGA processor-based designs.

What were the biggest challenges in designing a high-performance RISC-V SoC from scratch in India?

One of the biggest challenges was securing the right talent. Hiring experienced processor designers is difficult, as specialised expertise is limited in India. We built on prior experience from an earlier processor based on a different ISA, while training fresh graduates and collaborating with IITs to form the team. This issue is not limited to government projects; very few companies in India work deeply in this space. Many successful startups are led by Indian-origin professionals in Silicon Valley with 20-30 years of expertise at firms like Apple, Google, or Intel; expertise that is hard to find locally.

Fabrication was another major hurdle, since we had to rely mostly on external foundries. The first chip was fabricated at Silterra in Malaysia, the second at SCL Chandigarh (180nm), and the third at TSMC (28nm) in Taiwan. Initial tape-outs

were test or shuttle runs to validate designs. As a smaller player, securing slots with major foreign foundries was difficult because foundries prioritise large-volume customers, and procurement and coordination with foreign entities were challenging. Post-Covid supply constraints and memory shortages made access even harder. The situation is improving, however, with initiatives such as the India Semiconductor Mission and Tata's upcoming foundries expected to ease chip design and manufacturing in India.

How did you ensure verification completeness for such a complex SoC, and reach confidence that it is fully verified?

Verification followed a layered approach. It began at the module level, covering components like the pipeline and cache. After module-level verification, the modules were integrated to form the processor, where testing continued at the processor level. Initial simulations were run, followed by execution on an emulation platform. Once sufficient confidence was achieved, the design was implemented on an FPGA using a minimalistic SoC built around the processor. This SoC included basic infrastructure, an Ethernet interface for connectivity, a UART for terminal access, an SDRAM or DRAM interface for instructions and data, and program memory for verification.

Extensive FPGA testing was conducted using standard routines, publicly available tests, and third-party EDA vendor tools that generate a wide range of test routines. These tools supported long regression runs with self-checking mechanisms to detect failures. Once initial tests passed, more exhaustive verification was performed, covering different instruction classes, instruction types, combinations, and randomised scenarios. This iterative, multi-pronged verification process helped ensure both design robustness and confidence in processor reliability.

Can RISC-V help India build a full semiconductor stack, and what does that stack include?

Yes, RISC-V can help India build a full semiconductor stack, covering the entire value chain from chip design and IP to fabrication, testing, and product deployment. Today, it acts as a catalyst, supported by government initiatives like the Design Linked Incentive (DLI) scheme, which provides access to EDA tools, financial support for manpower, and subsidised IP. Many companies under the scheme have already started adopting RISC-V for the chips and devices they are developing. The key advantage is that RISC-V removes a major barrier to entry: unlike proprietary architectures, it eliminates upfront licensing fees and royalties, allowing Indian companies to design their own processors and SoCs more freely. This shift is already visible, with several Indian firms designing RISC-V-based processors and SoCs, though many are currently fabricated outside India.

Looking ahead, in the next 2-3 years, large-scale impact may remain limited, but over 5-10 years, significant developments are expected. Within 4-5 years, a more complete stack could emerge, including chips designed in India using largely indigenous IP, fabricated in domestic foundries such as Tata's upcoming facilities, followed by testing, reference design development, and production. Since Indian designs are already in development, these domestic foundries will have a ready customer base once operational. Over time, this ecosystem, driven by RISC-V and government support, could enable India to build a self-reliant, end-to-end semiconductor industry. **EFY**

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/IntJune26>

